

Suraj Kumar Sahoo

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Professional Summary

Software Developer specializing in **Agentic AI and Distributed Systems** with 1.5+ years of experience at the **National Informatics Centre (NIC)**, building production-grade multi-agent platforms and backend architectures for Government of India initiatives. Proven expertise in AgentScope, Model Context Protocol (MCP), gRPC streaming, self-validating Text-to-NoSQL query pipelines, and hybrid vector search over 1.3M+ records.

Technical Skills

Agentic AI & LLMs: Multi-Agent Orchestration, Model Context Protocol (MCP), Autonomous Tool Calling, Semantic Routing, Agentic Memory, Query Planning & Reflection, Hallucination Guardrails, RAG, AgentScope, LangGraph

Languages: Python (Asyncio), JavaScript, SQL

Backend & Distributed Systems: FastAPI, Node.js, Express.js, gRPC, Protocol Buffers (proto3), REST APIs, Microservices

Databases & Vector Stores: Qdrant Vector Store, MongoDB (Aggregation Pipelines, Motor), Redis Caching, SQLite

DevOps & Observability: Docker, Docker Compose, OpenTelemetry (OTel), Jaeger Tracing, Prometheus, Grafana, Git, GitLab

Professional Experience

National Informatics Centre (NIC)

Bhubaneswar, Odisha

Software Developer

Nov 2025 – Present

Project: SAINET – Smart Agriculture Intelligence Network

- Architected a production-grade **AgentScope-based Distributed Multi-Agent platform** coordinating a central Master Agent and 6+ specialized domain services using **gRPC proto3 streaming** and **Docker Compose**.
- Integrated a centralized **MongoMCP service (Model Context Protocol over HTTP/SSE)**, enabling autonomous agents to execute schema-safe query tools with deterministic execution.
- Built an agentic **Text-to-Mongo query compiler** coupled with an AST-level **QueryPlanValidator** to eliminate hallucinated queries and safely execute complex aggregations across **1.3M+ production records**.
- Implemented semantic entity resolution and agentic memory using **Qdrant vector store ('india_geomaster')**, enabling agents to autonomously resolve colloquial aliases and agricultural taxonomy.
- Evaluated system via a **50-query live benchmark**, achieving **100.0% factual precision (150/150 ground-truth facts)** with an **11.26s** P50 latency and zero runtime crashes across multi-season queries.
- Engineered distributed tracing across agent runtimes using **OpenTelemetry**, **Jaeger**, and **Prometheus**, streaming live agent telemetry and token chunks via **FastAPI Server-Sent Events (SSE)**.

AI Engineer Intern

Jan 2025 – Oct 2025

Project: BILLION-EYE – Government AI Initiative

- Developed scalable backend services using **Node.js**, **Express.js**, and **MongoDB** for real-time traffic incident analytics.
- Designed MongoDB aggregation pipelines, indexing strategies, and REST APIs for analytics dashboards, reducing query latency by **35%**.
- Optimized database performance through query tuning, indexing on large datasets, and Redis caching layers.
- Collaborated with the development team to enhance backend reliability, Docker containerization, and production readiness.

Project

DocumentSeeker – Enterprise Knowledge Retrieval

- Built an enterprise semantic document retrieval platform using **Qdrant Vector Database** and **Sentence Transformers**.
- Developed Retrieval-Augmented Generation (RAG) pipelines for intelligent document search, legal/regulatory policy Q&A, and strict citation grounding to prevent hallucinations.

Certification

Certified Python Developer Associate (CPDA-24)

Lifetime Credential

Education

Indira Gandhi Institute of Technology (IGIT), Sarang

2021 – 2025

B.Tech in Electronics & Telecommunication

CGPA: 7.83 / 10.0